

CLINICAL TRIALS AND OBSERVATIONS

CME Article

Blinatumomab consolidation for high-risk Ph⁻ B-cell acute lymphoblastic leukemia: the GRAALL-2014/B-QUEST study

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KEY POINTS

- Blinatumomab consolidation improves MRD clearance, and disease-free and overall survival in adults with HR Ph⁻ B-ALL.
- The benefit of additional alloHSCT for patients with poor MRD before blinatumomab consolidation remains to be defined.

Intensified chemotherapy regimens have improved outcomes in adults with Philadelphia chromosome-negative (Ph⁻) B-cell acute lymphoblastic leukemia (B-ALL), yet relapse remains a major cause of treatment failure and death. Blinatumomab is a T-cell engager that demonstrated marked activity in relapsed and measurable residual disease-positive (MRD⁺) B-ALL, as well as in frontline consolidation for MRD⁻ patients. The phase 2 GRAALL-2014/B-QUEST substudy evaluated the integration of blinatumomab into consolidation and maintenance therapy for adults with high-risk (HR) B-ALL. Between 2015 and 2020, 489 adults aged 18 to 59 years with newly diagnosed Ph⁻ B-ALL were enrolled, and among these 259 were classified as HR (presence of *KMT2A*-r, *IKZF1* deletion, or end-of-induction MRD $\geq 10^{-4}$). A total of 94 patients with HR were enrolled in the QUEST study (2018-2020) and received up to 5, 28-day cycles of blinatumomab during consolidation and maintenance. In addition, 90 patients with HR who were treated before QUEST activation without blinatumomab served as controls in a post hoc analysis.

Baseline characteristics were comparable between groups. Blinatumomab consolidation significantly improved MRD clearance, reduced relapse, and prolonged survival. At 5 years, cumulative incidence of relapse, disease-free survival (DFS), and overall survival were 23%, 68%, and 79% in the blinatumomab group compared with 49% ($P = .001$), 42% ($P = .001$), and 60% ($P = .03$) in controls. Patients eligible for allogeneic

hematopoietic stem cell transplantation (alloHSCT) derived overall benefit. However, no clear additional DFS advantage was observed among those who ultimately underwent transplantation. These findings support the frontline integration of blinatumomab and warrant prospective evaluation of transplantation strategies in this setting. This trial was registered at www.clinicaltrials.gov as #NCT03709719.



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Disclosures

CME questions author Laurie Barclay, freelance writer and reviewer, Medscape Education, declares no competing financial interests.

Learning objectives

Upon completion of this activity, participants will:

1. Describe overall outcomes with integration of blinatumomab into consolidation and maintenance therapy for adults with high-risk (HR) B-cell acute lymphoblastic leukemia (B-ALL), based on findings and post hoc analysis from the phase 2 GRAALL-2014/B-QUEST substudy
2. Determine outcomes with integration of blinatumomab into consolidation and maintenance therapy among various subgroups of adults with HR B-ALL, based on findings and post hoc analysis from the phase 2 GRAALL-2014/B-QUEST substudy
3. Identify clinical implications of outcomes with integration of blinatumomab into consolidation and maintenance therapy for adults with HR B-ALL, based on findings and post hoc analysis from the phase 2 GRAALL-2014/B-QUEST substudy

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Introduction

The management of adult Philadelphia chromosome-negative (Ph⁻) B-cell acute lymphoblastic leukemia (B-ALL) has greatly benefited from the intensification of chemotherapy regimens.¹⁻⁴ However, these benefits have been offset by an increase in toxicity, leading to treatment delays, dose reductions, and excessive adverse effects.⁵ Allogeneic hematopoietic stem cell transplantation (alloHSCT) reduces the risk of relapse and improves survival in treatment-resistant patients, albeit at the cost of additional toxicity.^{6,7} Among young adults aged ≤60 years, 5-year survival rates rarely exceed 70%. Progress with chemotherapy alone is now likely limited in adults, suggesting that a ceiling effect has been reached.

Blinatumomab is a bispecific antibody targeting CD19 on lymphoblasts and CD3 on T lymphocytes, enabling the redirection of

natural cytotoxicity toward cancer cells.⁸ After demonstrating its superiority over chemotherapy in relapsed/refractory (R/R) B-ALL, this immunotherapy was approved for B-ALL with detectable measurable residual disease (MRD).⁹⁻¹¹ More recently, the Eastern Cooperative Oncology Group (ECOG) E1910 phase 3 study demonstrated the benefit of blinatumomab-based consolidation for patients with a favorable MRD response after induction therapy, showing improved relapse-free and overall survival (OS) compared with chemotherapy-only consolidation.¹²

Following the Group for Research on Adult Acute Lymphoblastic Leukemia (GRAALL)-2005 protocol, we demonstrated that the independent prognostic factors associated with relapse risk for patients with Ph⁻ B-ALL include the presence of a *KMT2A* rearrangement (*KMT2A*-r), an intragenic *IKZF1* deletion (*IKZF1*del), or a positive MRD at the end of induction therapy.¹³ In the GRAALL-2014 protocol, we sought to

determine whether these high-risk (HR) patients might benefit from the addition of blinatumomab during consolidation.

In this study, we report the results of the phase 2 GRAALL-2014/B-QUEST study, which aimed at evaluating the benefit of adding blinatumomab during consolidation for patients with HR B-ALL. We took advantage of the delayed initiation of this nested study to compare the outcomes of these patients with those from the parent GRAALL-2014-B study who had HR features but were not exposed to blinatumomab.

Patients and methods

Study design

The GRAALL-2014/B-QUEST study was designed as a multicenter, open-label, phase 2 trial nested within the broader GRAALL-2014 protocol, previously published ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT02617004) identifier: NCT02617004).¹⁴ The GRAALL-2014 and GRAALL-2014/B-QUEST studies were conducted across 79 French, Belgian, and Swiss centers within the GRAALL network.

Study population

Between December 2015 and December 2020, the GRAALL-2014/B trial enrolled 489 adults aged 18 to 59 years with newly diagnosed Ph⁻ B-ALL, in the absence of evolving malignancy, pregnancy, HIV infection, active viral hepatitis, or any condition contraindicating intensive chemotherapy.¹⁴ Centralized diagnostic sampling was required at study entry. Patients were eligible for inclusion in the GRAALL-2014/B-QUEST substudy if they met the defined criteria, including (1) sustained complete remission (CR) at the start of the second consolidation phase (week 12); (2) no central nervous system (CNS) involvement at baseline; and (3) at least 1 HR feature, defined as the presence of KMT2A-r, IKZF1del, or positive MRD at the end of induction (MRD1 $\geq 10^{-4}$), including patients who required a salvage course to achieve CR. Notably, no predefined quantitative threshold of CD19 surface expression was required for inclusion in the QUEST substudy. Patients enrolled in the GRAALL-2014/B trial before initiation of the QUEST substudy and who met the same eligibility criteria but were not exposed to blinatumomab, were retrospectively included as an internal control cohort.

Diagnosis and response assessment

Initial diagnostic workup was performed locally and included cytomorphology, immunophenotyping, and cytogenetic analyses, including conventional karyotyping and fluorescence in situ hybridization, with mandatory screening for t(9;22)/BCR::ABL1 and KMT2A-r. Comprehensive molecular characterization of B-ALL cases was centralized at Saint-Louis University Hospital (Paris, France), including screening for IKZF1del and Ph-like genetic alterations.^{15,16} Treatment response to induction was assessed by cytomorphology on day 35. MRD was evaluated by immunoglobulin/T-cell receptor gene rearrangement quantitative polymerase chain reaction according to EuroMRD guidelines at 4 predefined time points such as after induction (MRD1), after first (MRD2) and second (MRD3) consolidation phases, and at the start of maintenance or at ~100 days after alloHSCT (MRD4). MRD testing was centralized in 6 EuroMRD-accredited laboratories.

Prednisone response was evaluated on day 8 of the steroid prephase and defined by a peripheral blast count below $1 \times 10^9/L$. CR required bone marrow blasts $<5\%$, absence of extramedullary disease, and hematologic recovery with platelet counts $>100 \times 10^9/L$ and an absolute neutrophil count $>1 \times 10^9/L$. Relapse was defined by the reappearance of $\geq 5\%$ blasts in the bone marrow or leukemic infiltration at any extramedullary site after prior CR.

Treatment schedule

The overall treatment backbone followed the age-adapted GRAALL-2014 protocol, previously described ([Figure 1](#); supplemental Table 1, available on the *Blood* website).¹⁴ Briefly, all patients received a Berlin-Frankfurt-Münster-like induction regimen, followed, if CR was not achieved, by a high-dose cytarabine-based salvage course. Patients attaining CR proceeded to 3 consolidation blocks, with a delayed intensification phase between consolidations 2 and 3, followed by a 2-year maintenance phase consisting of monthly pulses of vincristine and prednisone, along with continuous oral 6-mercaptopurine and methotrexate therapy. Each consolidation cycle included 3 blocks consisting of high-dose cytarabine, high-dose methotrexate, and cyclophosphamide, respectively. In the QUEST substudy, blinatumomab was incorporated into both consolidation and maintenance phases. Patients received 5, 28-day cycles of continuous IV blinatumomab at 28 $\mu g/day$. The first 2 cycles of blinatumomab replaced the cyclophosphamide blocks in consolidation 2 and 3. The remaining 3 cycles were administered at months 1, 3, and 5 of maintenance, substituting for standard pulses of vincristine and prednisone, along with continuous oral 6-mercaptopurine and methotrexate therapy. Patients with very HR (VHR) disease were eligible for alloHSCT in case of poor early response to treatment, defined by MRD1 $\geq 10^{-3}$ (including patients who required a salvage course to achieve CR) or MRD2 $\geq 10^{-4}$. For patients planned to receive alloHSCT, blinatumomab could be continuously given until the start of the conditioning regimen. For patients not included in the QUEST study, transplantation was scheduled after the second consolidation course.

Statistical methods

Sample size justification The primary objective was to demonstrate, in a noncontrolled study, that the 3-year disease-free survival (DFS) rate in the HR population was improved from the historical reference of 50% to 65%, with a requirement of 95 patients, a 1-sided type I error rate of 0.05, and a statistical power of 0.90.

Statistical analysis In addition to the primary efficacy analysis, a post hoc comparison was conducted between patients in the QUEST cohort and an internal control cohort who met identical eligibility criteria but were not exposed to blinatumomab. Binary outcomes were compared using Fisher exact test, and continuous variables were analyzed using the Wilcoxon rank-sum test. Time-to-event end points, including OS and DFS, were calculated from the start of consolidation 2 to the event or last follow-up and compared using the log-rank test. The cumulative incidence of relapse (CIR) was estimated in the presence of competing risks, with death in CR treated as a competing event; comparisons were performed using the Gray test.¹⁷ To quantify the effect of blinatumomab exposure on outcomes, Cox

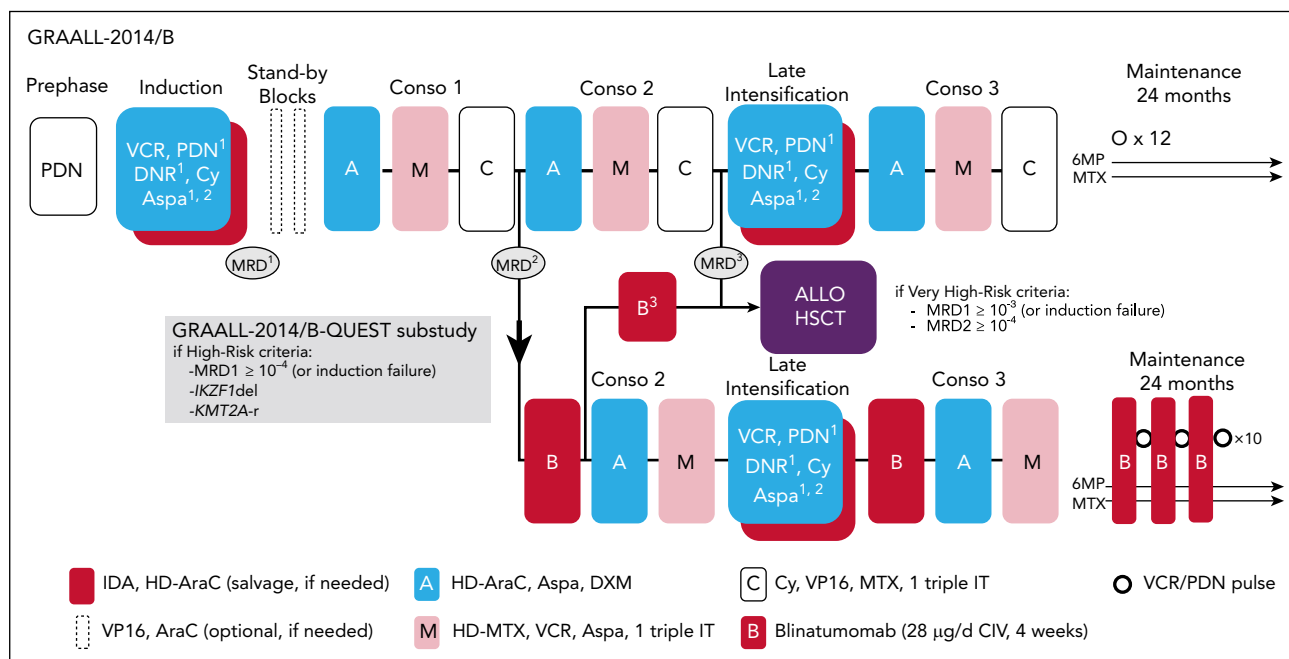


Figure 1. GRAALL-2014/B and QUEST substudy schemes. ¹Age-adapted doses for patients aged >45 years. ^{14,2}L-Asparaginase frontline, switch to crisantaspase if allergy, low activity, or antibody is detected. ³Blinatumomab could be given continuously until alloHSCT (a total of 28 days exposure was recommended). AraC, cytarabine; Aspa, L-asparaginase; CIV, continuous IV infusion; Conso, consolidation; Cy, cyclophosphamide; DNR, daunorubicin; DXM, dexamethasone; HD, high-dose; IDA, idarubicin; IT, intrathecal; MTX, methotrexate; 6MP, 6-mercaptopurine; PDN, prednisone; TBI, total body irradiation; VCR, vincristine; VP16, etoposide.

proportional hazards models were fitted, with the proportional hazards assumption assessed by inspection of Schoenfeld residuals and the Grambsch-Therneau test.^{18,19} As a sensitivity analyses, multivariable Cox proportional hazards models (DFS, OS) and Fine-Gray subdistribution hazard models (CIR) were fitted including age, sex, log₁₀ white blood cell count (WBC), VHR status, and treatment cohort. Analyses were performed using Stata statistical software version 18.0 (College Station, TX).

The trial was designed and reviewed by the GRAALL scientific committee and approved by the French Ethical Committee Review Board (Comité de Protection de la Personne) Ile-de-France VI (number 2014/66). In accordance with the Declaration of Helsinki, informed consent was obtained before study entry.

Results

Patient selection

Between December 2015 and December 2020, the GRAALL-2014/B trial enrolled 489 patients, of whom 455 (93.1%) achieved CR (Figure 2). Among patients in CR, 259 (56.9%) met HR criteria, 173 (38.0%) were classified as standard risk, and 23 (5.1%) could not be classified. Among 259 patients who were HR, 128 (49.4%) were included before QUEST activation and 131 (50.6%) were included after QUEST activation. To illustrate the impact of this risk stratification in the preblinatumomab era, outcomes of the 128 HR and the 85 standard-risk patients treated in the GRAALL-2014/B before QUEST activation are shown in the supplemental Figure 1.

Among the 259 patients who were HR, 215 (83.0%) proceeded to the start of consolidation 2 (Figure 2). The most frequent

reason for not starting consolidation 2 was early exposure to blinatumomab, as per investigator decision (n = 17). Other reasons are summarized in Figure 2. Among the 215 patients starting consolidation 2, 12 had baseline CNS involvement. Of the remaining 203 potentially eligible patients, 94 were enrolled in the QUEST substudy between October 2018 and December 2020, and 90 had been treated before QUEST activation and served as the internal control cohort. After QUEST trial activation, 19 patients were not included, including 13 because of investigator decision and 6 because the substudy had closed by the time they reached consolidation 2 (Figure 2).

Therefore, the evaluable QUEST cohort comprised 94 patients, and the internal control cohort included 90 patients consecutively accrued in the GRAALL-2014/B trial who met the same HR eligibility criteria but not treated with blinatumomab. Of note, 1 additional patient with CD19⁺ T-ALL was erroneously enrolled in the QUEST cohort and was excluded from the final analysis (95 patients included).

Patient characteristics

Baseline characteristics of patients in the QUEST and control cohorts are summarized in Table 1. No significant differences were observed between the 2 groups in terms of demographic features, initial disease presentation, early treatment response, or risk stratification criteria. Median age was 34 years (range, 18-59) in the QUEST cohort and 35 years (range, 18-59) in the control cohort (P = .82). Median WBC count at diagnosis was 12 × 10⁹/L (range, 1-449) and 7 × 10⁹/L (range, 0-637), respectively (P = .36). As per the eligibility criteria, no patients had CNS involvement at baseline, and all achieved CR, including 5 (5%) after salvage course in QUEST and 8 (9%) in the control cohort (P = .40).

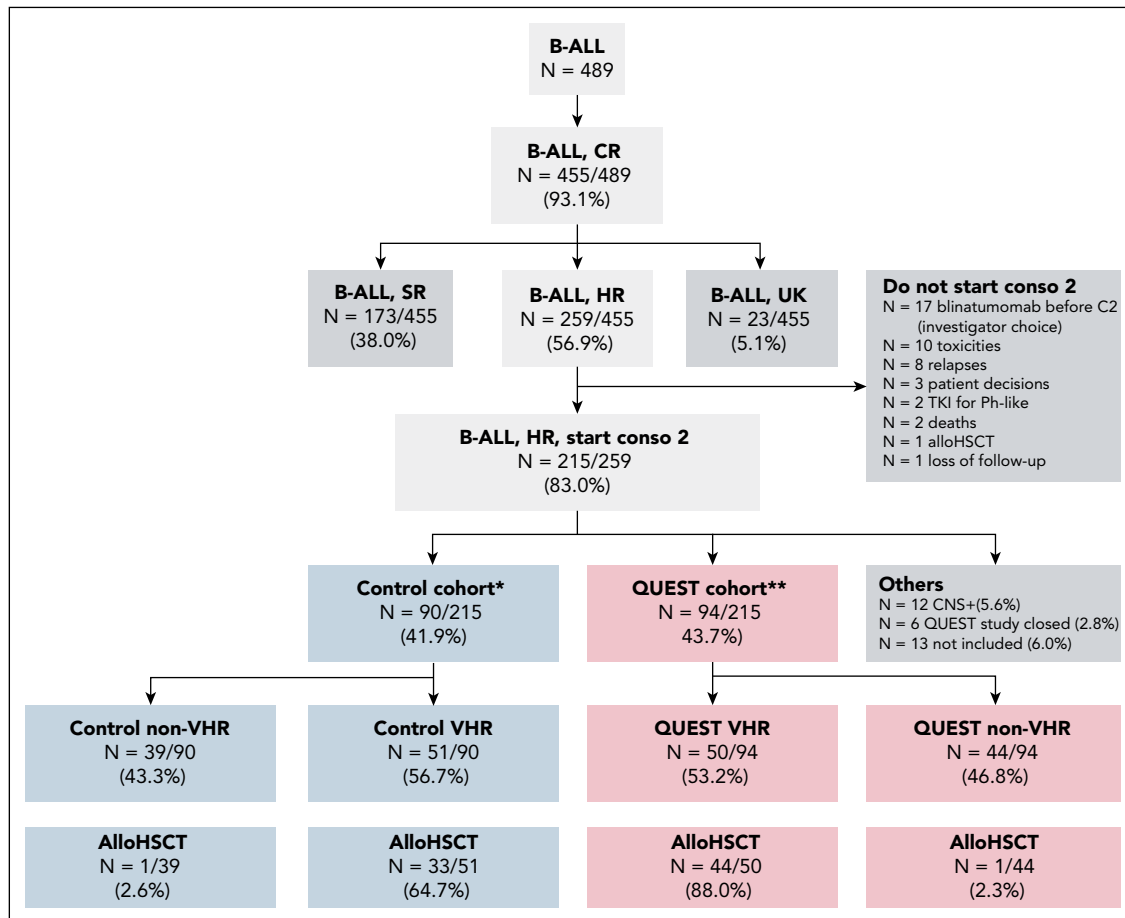


Figure 2. Patient flowchart. *All patients meeting QUEST criteria before QUEST activation. **One additional patient with CD19⁺ T-ALL excluded from the analysis. SR, standard risk; TKI, tyrosine kinase inhibitor; UK, unknown subtype.

Regarding HR features, the frequency of *KMT2A-r* was 17% in the QUEST cohort and 23% in the control group ($P = .36$), *IKZF1del* was present in 40% and 36% of patients, respectively ($P = .65$), and MRD1 was $\geq 10^{-4}$ in 73% of QUEST patients vs 80% in controls ($P = .29$). The frequency of other genetic subtypes did not differ significantly between cohorts (Table 1; supplemental Table 2). At the end of the first consolidation, the rate of patients with MRD2 $\geq 10^{-4}$ remained comparable (45% vs 39%, respectively; $P = .54$). The rate of patients eligible for alloHSCt was 53% (50/94) in the QUEST cohort and 57% (51/90) in the control group (Figure 2).

Patient outcomes

After consolidation 2, the rate of undetectable MRD3 was 72% (62/86) with blinatumomab and 55% (37/67) after chemotherapy alone ($P = .041$; Table 2). Among patients with preconsolidation 2 MRD2 $\geq 10^{-4}$, the rate of conversion to undetectable MRD3 was 51% (20/39) after blinatumomab and 17% (4/24) after chemotherapy ($P = .008$; Table 2).

As anticipated, median follow-up duration was significantly shorter in the QUEST cohort (5.3 years) compared with the control cohort (7.0 years; $P < .001$). Among patients eligible for alloHSCt, 44 of 50 (88%) were transplanted in the QUEST study, whereas only 33 of 51 (65%) underwent transplantation in the control group ($P = .004$; Table 2). In both cohorts, 1

patient received alloHSCt outside of formal indication after investigator decision (Table 2). The protocol did not prospectively record the specific reasons for not proceeding to alloHSCt in individual cases.

The total number of relapses was lower in the QUEST cohort ($n = 22$) than in the control cohort ($n = 46$). In the QUEST arm, relapses included 16 bone marrow, 3 isolated CNS events, 1 combined (bone marrow + CNS), and 2 isolated extramedullary relapses. In the control group, there were 38 bone marrow relapses, 2 isolated CNS, 3 combined, and 3 other isolated extramedullary relapses.

The primary outcome of the phase 2 study was reached with a 3-year DFS of 70% (95% confidence interval [CI], 60-78), with a lower bound above the historical 50%. Of note, the 3-year DFS in the control cohort was 48% (95% CI, 38-58), which substantiated this 50% hypothesis.

At 5 years, the CIR, DFS, and OS in the QUEST cohort were 23% (95% CI, 16-33), 68% (95% CI, 58-76), and 79% (95% CI, 69-86), respectively (Table 2; Figure 3). In contrast, corresponding 5-year outcome estimates in the control group were 49% (95% CI, 40-60) for CIR, 42% (95% CI, 31-52) for DFS, and 60% (95% CI, 49-70) for OS. Although based on a nonrandomized comparison, these results suggest a significant benefit of

Table 1. Patient, disease, and early response characteristics

	QUEST (N = 94)	Control (N = 90)	P value
Median age, y (range)	34 (18-59)	35 (18-59)	.82
Male/female	51/43	40/50	.19
Median WBC, $\times 10^9/L$ (range)	12 (1-449)	7 (0-637)	.36
CNS involvement, n	0	0	1.00
Genetics, n (%)			
KMT2A-r*	16/94 (17)	21/90 (23)	.36
Ph-like	17/94 (18)	16/90 (18)	1.00
DUX4-r	13/94 (14)	10/90 (11)	.66
ZNF384-r	10/94 (11)	5/90 (6)	.28
Low hypodiploidy/near triploidy	8/94 (9)	6/90 (7)	.78
CDX2/UBTF	4/94 (4)	7/90 (8)	.36
Other (<5%)	19/94 (20)	19/90 (21)	1.00
Not classified	2/94 (2)	2/90 (2)	1.00
Not evaluable	5/94 (5)	4/90 (4)	1.00
IKZF1del*	37/93 (40)	32/88 (36)	.65
Early outcome†			
CR (N) %	94 (100)	90 (100)	1.00
CR after salvage (N) %	5 (5)	8 (9)	.40
MRD1 $\geq 10^{-4}$ * (N) %	68/93 (73)	70/87 (80)	.29
MRD2 $\geq 10^{-4}$ (N) %	41/92 (45)	33/84 (39)	.54

See supplemental Table 2 for a full genetic description.

MRD1, after induction; MRD2, after consolidation 1.

*Criteria for HR stratification.

†Before inclusion in the QUEST substudy.

incorporating blinatumomab into frontline therapy, with a reduced risk of relapse (subdistribution hazard ratio [SHR] 0.41; 95% CI, 0.25-0.68), and improvements in both DFS (hazard ratio [HR] 0.48; 95% CI, 0.31-0.75) and OS (HR 0.54, 95% CI, 0.32-0.94). In multivariable analyses adjusting for baseline

covariates (age, sex, $\log_{10}$ WBC, and VHR status), blinatumomab exposure remained independently associated with improved DFS (HR, 0.47; 95% CI, 0.30-0.74), reduced CIR (SHR, 0.38; 95% CI, 0.23-0.64), and improved OS (HR, 0.50; 95% CI, 0.29-0.86) (supplemental Table 4).

Table 2. Patient outcomes

	QUEST (N = 94)	Control (N = 90)	P value
MRD3 status			
Undetectable (%)	62/86 (72)	37/67 (55)	.041
Undetectable if MRD2 $\geq 10^{-4}$ (%)	20/39 (51)	4/24 (17)	.008
Median follow-up, y	5.3	7.0	<.001
AlloHSCT, (%)			
VHR	50/94 (53)	51/90 (57)	.66
AlloHSCT rate	45/94 (48)	34/90 (38)	.18
AlloHSCT rate among VHR	44/50 (88)	33/51 (65)	.009
Long-term outcome, % (95% CI)			
5-year CIR	23 (16-33)	49 (40-60)	.001
5-year DFS	68 (58-76)	42 (31-52)	.001
5-year OS	79 (69-86)	60 (49-70)	.03

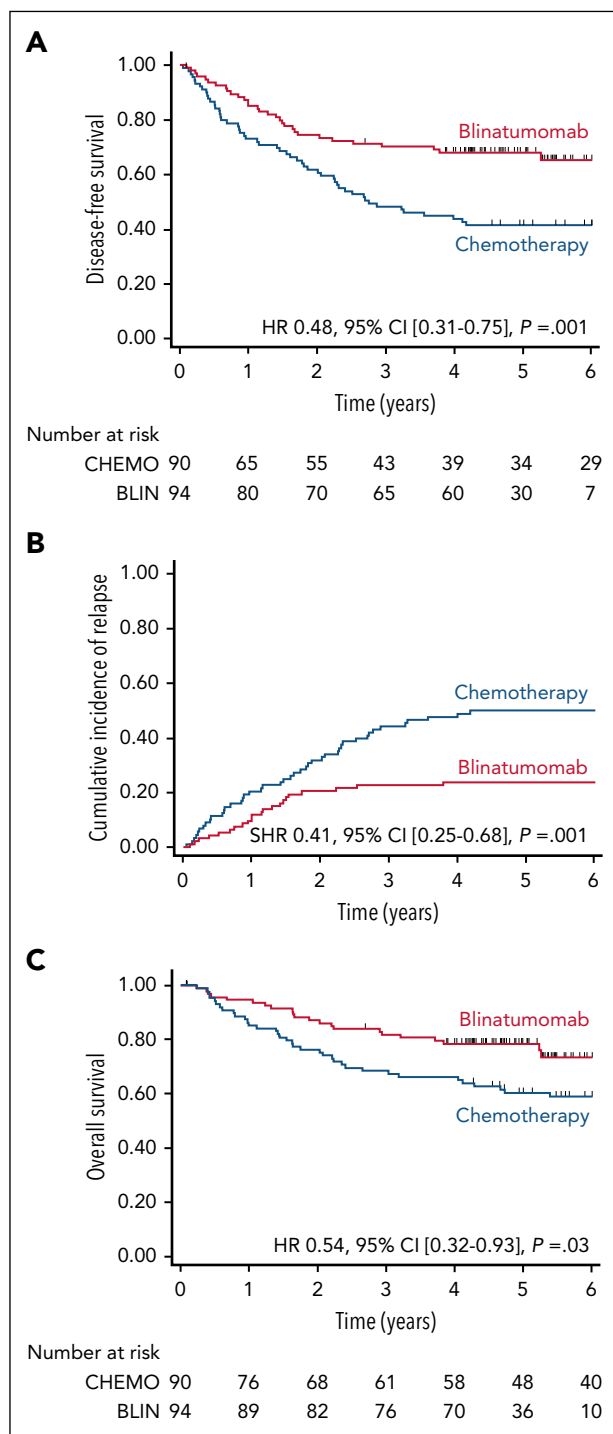


Figure 3. Patient outcomes. (A) DFS; (B) CIR; (C) OS in both cohorts. BLIN, blinatumomab; CHEMO, chemotherapy.

The benefit of blinatumomab across patient subgroups, particularly those defined by HR criteria (*KMT2A-r*, *IKZF1del*, or $MRD1 \geq 10^{-4}$) is illustrated in Figures 4 and 5. Overall, blinatumomab consolidation was consistently associated with a significant reduction in CIR and improvement in DFS compared with chemotherapy alone, regardless of subgroup (Figure 4). Among the 3 defining HR features, patients with $MRD1 \geq 10^{-4}$ or *IKZF1del* derived clear benefit from blinatumomab, showing both lower relapse incidence and superior DFS (Figure 5). In

contrast, no statistically significant benefit was observed in the *KMT2A-r* subgroup. Patients with Ph-like B-ALL also benefited from blinatumomab in terms of CIR (SHR, 0.24; 95% CI, 0.07-0.89) and DFS (HR, 0.34; 95% CI, 0.11-0.99) (Figures 4 and 5; supplemental Figure 3). A benefit in OS was observed for patients aged ≥ 45 years, if WBC counts were $\geq 30 \times 10^9/L$, or in case of *IKZF1del* (supplemental Figure 2).

Patients eligible to alloHSCT (VHR)

To investigate the imbalance in transplant rates observed among patients who were VHR, we compared time to transplant, incidence of early relapse, and donor profiles between the 2 cohorts. The median time from the start of consolidation 2 to transplantation was similar in both the QUEST and control cohorts, 71 days (range, 36-226) vs 73 days (range, 41-188), respectively ($P = .53$). Within the first 120 days after initiation of consolidation 2, early relapses occurred more frequently in the control cohort ($n = 4$) compared with the QUEST cohort ($n = 1$), potentially limiting access to transplant. Although the use of haploidentical donors was not recommended per protocol, such donors were used in 8 patients in the QUEST cohort vs 2 in the control cohort. Other donor types were evenly distributed across both groups (supplemental Table 3).

Patients eligible for alloHSCT derived significant benefit from blinatumomab, with improved DFS (HR, 0.54; 95% CI, 0.31-0.95; Figures 4 and 6) and a lower CIR (SHR, 0.41; 95% CI, 0.21-0.82; Figure 4). However, among patients who ultimately underwent transplantation, prior exposure to blinatumomab was not associated with a significant DFS advantage (HR, 0.76; 95% CI, 0.39-1.50; Figure 6). Notably, and consistent with the findings in the whole population (Table 2), among transplanted patients with available MRD3 assessment, MRD3 undetectable status was more frequent in the QUEST cohort than in controls (23/41 vs 8/27; 56% vs 30%; $P = .047$). Outcomes in the small subgroup of patients who were VHR and did not undergo alloHSCT, should be interpreted descriptively, given the limited sample size.

Adherence to blinatumomab schedule

The median number of blinatumomab cycles was 3 (range, 1-7) in the overall population ($n = 94$), 5 (range, 1-6) among patients not bridged to alloHSCT, and 2 (range, 1-7) among those bridged to alloHSCT ($n = 45$) (supplemental Figure 4). Among the 49 patients not bridged to alloHSCT, 10 (20%) received < 5 cycles, including 4 who received 1 cycle, 4 who received 2 cycles, and 1 who received 4 cycles. Reasons for early discontinuation included disease relapse ($n = 4$), neurotoxicity ($n = 3$; 2 grade 4, 1 grade 3), patient decision ($n = 2$), and MRD progression ($n = 1$).

Discussion

Over the past decade, mounting evidence has established blinatumomab as an effective component of frontline consolidation therapy for Ph⁺ ALL. The randomized ECOG-ACRIN E1910 trial provided the first controlled proof of this benefit for patients aged 50 to 70 years who achieved CR with MRD negativity after induction.¹² The GRAALL-2014/B-QUEST study broadens this evidence base by focusing on a HR population, including patients with suboptimal MRD response to

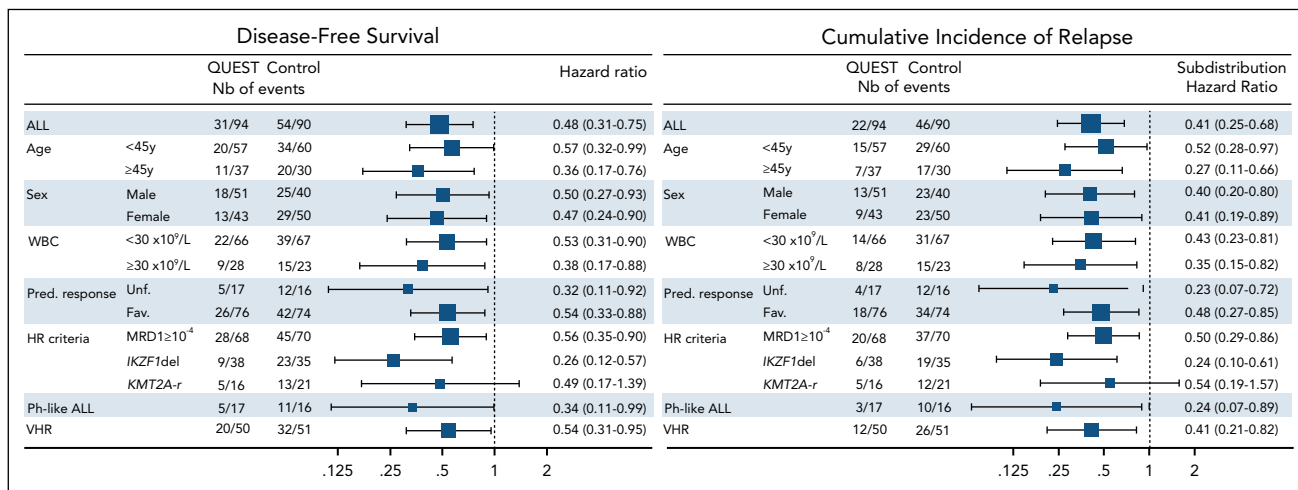


Figure 4. Subgroup analysis for DFS and CIR. Fav., favorable; Nb, number; Pred., prednisone; Unf., unfavorable.

pediatric-inspired regimen. By leveraging an internal control cohort treated on the same protocol before QUEST activation, the present study shows a survival advantage with blinatumomab consolidation.

After blinatumomab was approved for R/R B-ALL and for MRD-positive B-ALL, several studies investigated the role of blinatumomab in frontline consolidation for Ph⁺ ALL. The MD Anderson Cancer Center administered 4 cycles of blinatumomab consolidation after 4 cycles of hyper-CVAD (hyperfractionated cyclophosphamide, vincristine, doxorubicin, and dexamethasone), followed by a maintenance including 3 additional cycles of blinatumomab.²⁰ Blinatumomab consolidation was given after 2 hyper-CVAD courses for patients who are HR, defined by the persistence of MRD, the presence of KMT2A-r, Ph-like profile, TP53 mutation, hypodiploid, or near-tetraploid karyotype. In a cohort of 36 patients aged 17 to 59 years, the 3-year OS was 81%. The Italian GIMEMA (Gruppo Italiano Malattie Ematologiche Dell'Adulto) group recently published the results of the LAL2317 phase 2 study that evaluated the benefit of 2 cycles of blinatumomab consolidation in 149 patients aged 18 to 65 years.²¹ Blinatumomab started after 2 chemotherapy consolidation courses (week 10), with 122 patients receiving at least 1 blinatumomab course (82%). The 3-year OS was 71%, reaching 77% in younger patients aged 18 to 40 years. The E1910 study included 488 patients aged 30 to 70 years and randomized 224 patients who achieved a complete MRD response (below 10⁻⁴) after 2 induction and 1 intensification cycles (week 14).¹² The investigational arm included 4 cycles of blinatumomab and 3 cycles of chemotherapy, whereas the standard-of-care arm included 4 chemotherapy cycles. For patients with MRD negative, the 3-year OS was 85% after blinatumomab vs 68% after standard-of-care treatment (HR, 0.41; 95% CI, 0.23 - 0.73; *P* = .002). The results of the present study are thus in line with prior publications, recognizing that it focused on patients who were HR. The optimal number of consolidation cycles of blinatumomab for adult Ph⁺ ALL remains unclear and to be further explored, also considering the benefit reported with shorter exposure in recent frontline pediatric and infant trials.^{22,23}

In our study, a consistent benefit of blinatumomab over chemotherapy was observed across analyzed subgroups. Notably, both younger and older adults derived similar benefit, in contrast to the Eastern Cooperative Oncology Group-American College of Radiology Imaging Network (ECOG-ACRIN) E1910 trial, where efficacy appeared attenuated in older patients.¹² In contrast, patients with KMT2A rearrangements did not experience a significant improvement in outcome. Although limited number of such cases (*n* = 37) precludes definitive conclusions, these findings are consistent with prior reports suggesting that KMT2A-rearranged B-ALL may be less sensitive to CD19-directed therapies with a higher propensity for immune escape through CD19 antigen loss, sometimes accompanied by lineage switch toward a myeloid phenotype.²⁴ The impact of blinatumomab in Ph-like B-ALL remains controversial. In the GIMEMA LAL2317 study, it was observed that blinatumomab did not abrogate the adverse prognostic impact of Ph-like genotype, as DFS remained inferior in this subgroup (*P* = .003).²¹ Conversely, in the R/R setting, some reports have suggested a higher likelihood of initial response in Ph-like ALL. In the present study, blinatumomab appeared to partially overcome the poor prognosis of Ph-like ALL, supporting future prospective evaluation of combinations with kinase inhibitors targeting the underlying signaling abnormalities.²⁵ In the GRAALL-2014 study, Ph-like ALL, frequently associated with IKZF1del and poor end-of-induction MRD, were almost always classified in the HR group (*N* = 53/59, 90%). These patients benefited from the adjunction of blinatumomab in consolidation. Notably, the subgroup that displayed the most marked improvement in relapse risk reduction and DFS improvement was the one with IKZF1del.

One of the major questions which emerge from the use of blinatumomab in consolidation and after the strong reduction in relapse risk is about the necessity to consolidate patients with alloHSCT. In the MD Anderson Cancer Center, 13 patients (34%) received alloHSCT in first CR, the vast majority for HR genetic features.²⁰ In the GIMEMA LAL2317 study, patients eligible for alloHSCT had either a high WBC

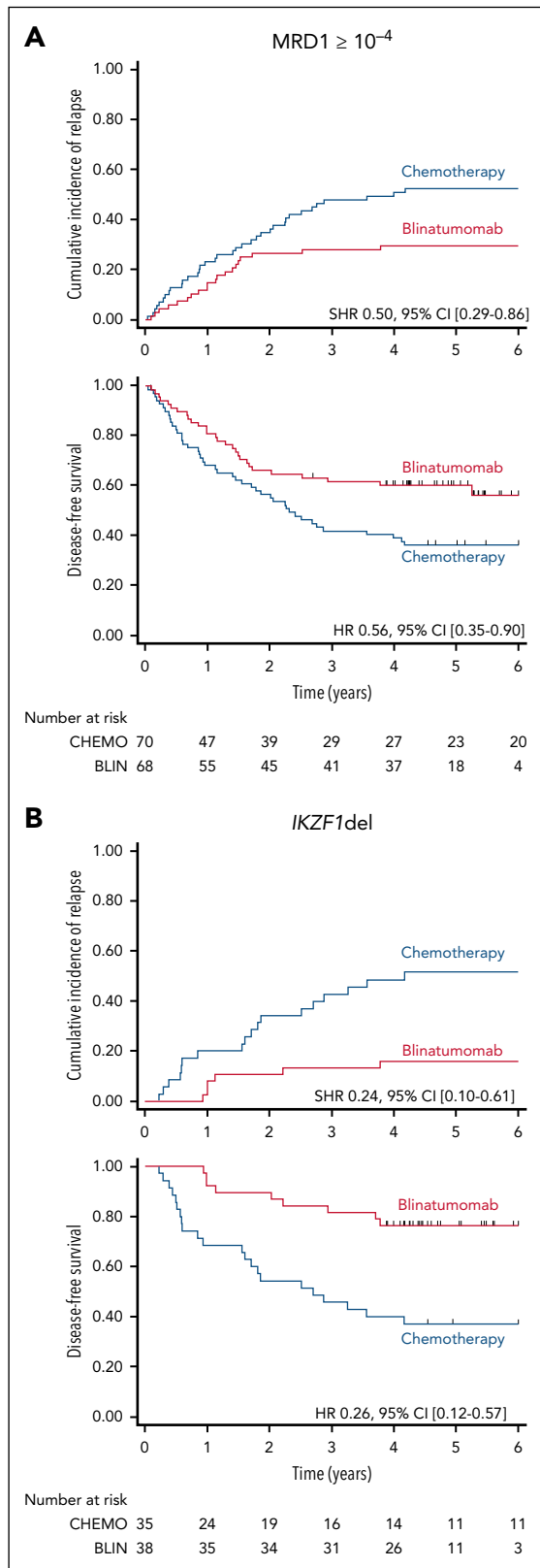


Figure 5. Outcome according to HR criteria. CIR and DFS in both cohorts for patients with (A) $MRD1 \geq 10^{-4}$, (B) $IKZF1del$, or (C) $KMT2A-r$.

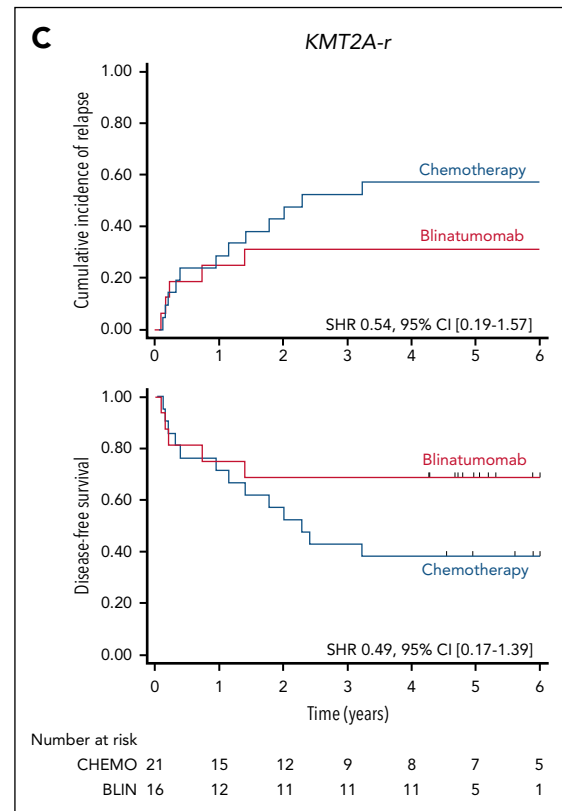


Figure 5 (continued)

($>100 \times 10^9/L$) or a HR cytogenetics ($KMT2A-r$, $+8$, -7 , $del6q$, $t[8;14]$), hypodiploidy/near triploidy, or complex karyotype with ≥ 5 abnormalities).²¹ Although this subgroup had an inferior DFS ($P = .0015$), a time-dependent analysis showed that they benefited from alloHSCT ($P = .0457$). In the E1910 study, indication for alloHSCT were defined as per investigator decision.¹² The rate of alloHSCT was well balanced between the 2 arms, also probably because all patients had reached MRD negativity at the time of inclusion. A landmark analysis which explored the role of alloHSCT in the blinatumomab arm did not suggest any benefit of alloHSCT, even when focusing on 50 patients with HR genetics.²⁶ In the present study, blinatumomab consolidation was associated with improved outcome for patients with poor MRD response who were eligible for alloHSCT. However, among patients who ultimately underwent transplantation, we did not observe a clear additional DFS advantage associated with prior blinatumomab exposure. Interpretation of this finding is limited by the nonrandomized design, differences in transplant rates between cohorts, and the small number of nontransplanted patients who are VHR. In addition, patients proceeding to alloHSCT received fewer blinatumomab cycles before conditioning, and a potential exposure effect cannot be excluded. Overall, these observations suggest that the role and timing of alloHSCT in the context of blinatumomab-based consolidation warrant prospective evaluation.

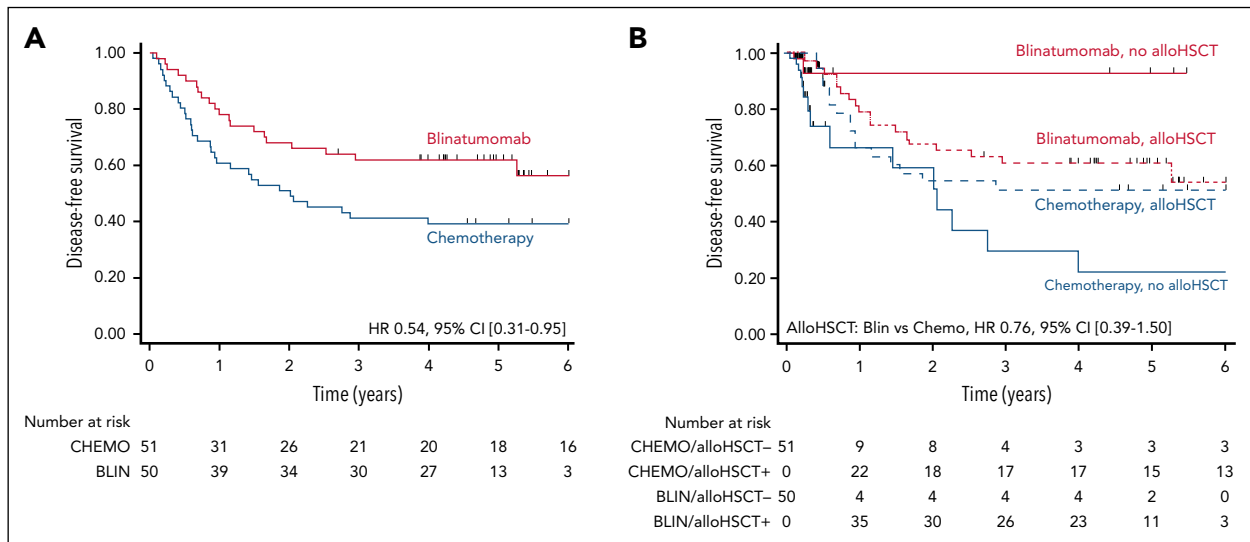


Figure 6. Outcome of patients eligible to alloHSCT. DFS for patients eligible according to (A) exposure to blinatumomab or not, (B) exposure to blinatumomab and/or alloHSCT (time-dependent analysis). AlloHSCT was treated as a time-dependent variable using a Simon-Makuch approach. All patients were classified as no alloHSCT at time zero and moved to the alloHSCT group at the time of transplantation.

The design of the present study and of the post hoc analysis warrant discussion. QUEST was a phase 2 trial aiming to demonstrate improved DFS with blinatumomab, targeting a 3-year DFS rate above the historical 50% benchmark observed in the GRAALL-2005 study. The post hoc analysis is a non-randomized comparison within the same parent protocol, including consecutively treated patients who met identical eligibility criteria most notably, achievement of sustained remission and initiation of consolidation 2. The 2 cohorts were well balanced for baseline demographics, disease characteristics, and early response, supporting the decision to forgo statistical matching or other adjustment methods. However, similarly to the E1910 study,¹² the design inherently introduces an immortal time bias, as patients receiving blinatumomab at week 12 had to remain alive and in continuous remission without prohibitive toxicity, to be eligible for inclusion in the substudy.

In conclusion, our study shows a valuable benefit of blinatumomab in a HR population of patients. Although the benefit of blinatumomab is observed in almost all subgroups, this study did not detect a benefit of prior blinatumomab treatment in the specific subgroup of patients who were bridged to alloHSCT because of a poor MRD response to chemotherapy. Based on the outcome improvements in Ph⁻ B-ALL and the outstanding results reported in Ph⁺ ALL with chemotherapy-free approaches, subsequent studies on Ph⁻ B-ALL should further explore the de-escalation of chemotherapy intensity and/or indication for alloHSCT. In the current GRAALL-2024 study, patients with unfavorable genetics or poor response to induction and who achieve a complete MRD response to blinatumomab, are randomized to receive alloHSCT or a combination of chemotherapy and blinatumomab only.

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Authorship

Contribution: V.L. provided administrative support; N.B., F.H., T.L., M.H., R.K., Y.H., M.P., P.C., M.B., C.P., E.D., S.M., A.T.-B., F.V.O., T.C., M.E.-B., N.S., J.K., A.B., A.V., F.P., I.V., L.S., S.B., M.A., M.L., L.F., C.B., N.I., C.G., Y.C., E.C., and H.D. provided study materials or patients; R.K., C.P., E.D., and E.C. performed central review of molecular and cytogenetic data; N.B. and H.D. performed data analysis and interpretation, and wrote the manuscript; and all authors provided final approval of the manuscript.

Conflict-of-interest disclosure: N.B. and F.H. received honoraria from Amgen, Novartis, Pfizer, Servier, and Gilead. H.D. received honoraria from Amgen and Servier. M.H. received honoraria from Novartis, Sanofi, Immedica, and Pierre Fabre and travel support from Medac, Sanofi, Novartis, and Pierre Fabre. P.C. received honoraria from Takeda, Servier, AbbVie, and Priothera and travel support from AbbVie. T.L. received honoraria from Servier and Amgen. The remaining authors declare no competing financial interests.

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Footnotes

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